

Simple Nonlinear Graphs



Curriculum Ready
ACMNA: 296, 239, 267



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SIMPLE NONLINEAR GRAPHS

Linear equations have the form $y = mx + c$ where the power of x (and y) is always 1. They are called 'Linear' because their graphs are straight lines. If the power is higher than 1, like $y = x^2 + 1$ or $y = x^3$ then the equations are not linear or "nonlinear". This books will show how to draw these types of graphs.

Answer these questions, *before* working through the chapter.

I used to think:

Which of these graphs is linear: **a** $y = \frac{x}{2}$ or **b** $y = \frac{2}{x}$? Why?

How is a quadratic equation different to a linear equation?

Which method can always be used to draw any graph for any equation?

Answer these questions, *after* working through the chapter.

But now I think:

Which of these graphs is linear: **a** $y = \frac{x}{2}$ or **b** $y = \frac{2}{x}$? Why?

How is a quadratic equation different to a linear equation?

Which method can always be used to draw any graph for any equation?



What do I know now that I didn't know before?



What is a Linear Graph?

A linear equation is an equation where the highest powers of both x and y are 1. A linear graph is based on a linear function like $y = 2x + 2$ or $y = -4x + 1$. In each case, the powers of x and y are both 1.

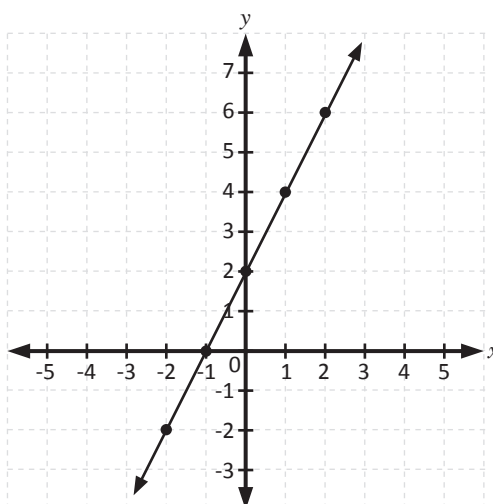
Using a Table to Plot a Graph

There are many methods that can be used to draw the graph of any function (linear or nonlinear) but a method that never fails, for any function (linear or otherwise) uses a table to plot points (x, y) on a coordinate plane.

Complete this table of values for $y = 2x + 2$ and use it to plot the graph on the grid below

x	-2	-1	0	1	2
y	-2	0	2	4	6

Now plot the points (x, y) : $(-2, -2)$, $(-1, 0)$, $(0, 2)$, $(1, 4)$, $(2, 6)$ and join the dots.



And of course the straight line goes through each point.

The table method works for any function, not only linear functions as above. This chapter will show how the table method can be used to draw:

- A Parabola. This is the graph of an equation where the power of x is 2.
For example $y = 3x^2$.
- A Hyperbola. This is the graph of an equation in the form $y = \frac{k}{x}$ where k is any number.
For example $y = \frac{12}{x}$.
- An Exponential. This is the graph of an equation which has x as the index of a number.
For example $y = 3^x$.

1. Identify which of these relationships is linear.

a $y = x + 2$

b $y = x^2 + 2x$

c $y = -3x + 3$

d $x + 2y = 6$

e $y = \frac{x}{2}$

f $y = \frac{2}{x}$

2. Complete the tables for these equations.

a $y = x + 3$

x	-2	-1	0	1	2
y					

b $y = 2x$

x	-2	-1	0	1	2
y					

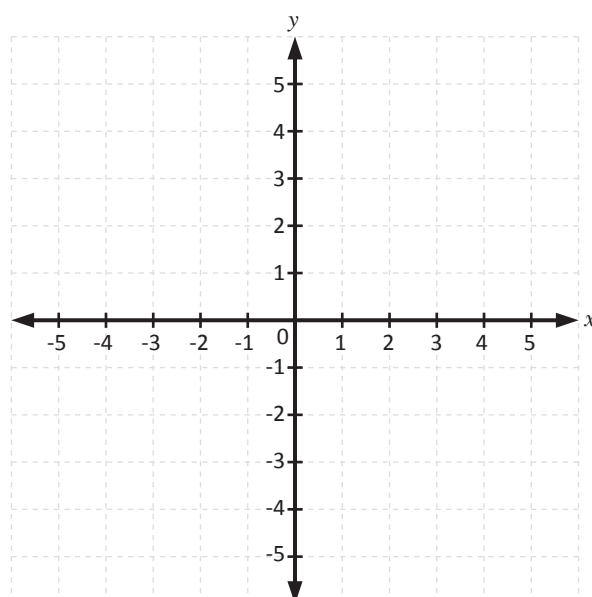
c $y = -x$

x	-2	-1	0	1	2
y					

d $y = -x - 3$

x	-2	-1	0	1	2
y					

3. Use the table d in the previous question to plot the line $y = -x - 3$ on the coordinate plane below.



What if the Highest Power is 2?

In the equation $y = ax^2$, (where a can be any number) the highest power of x is 2. The equation is not linear, but is called quadratic. The graph of a quadratic equation does not form a straight line, but a shape called a parabola.

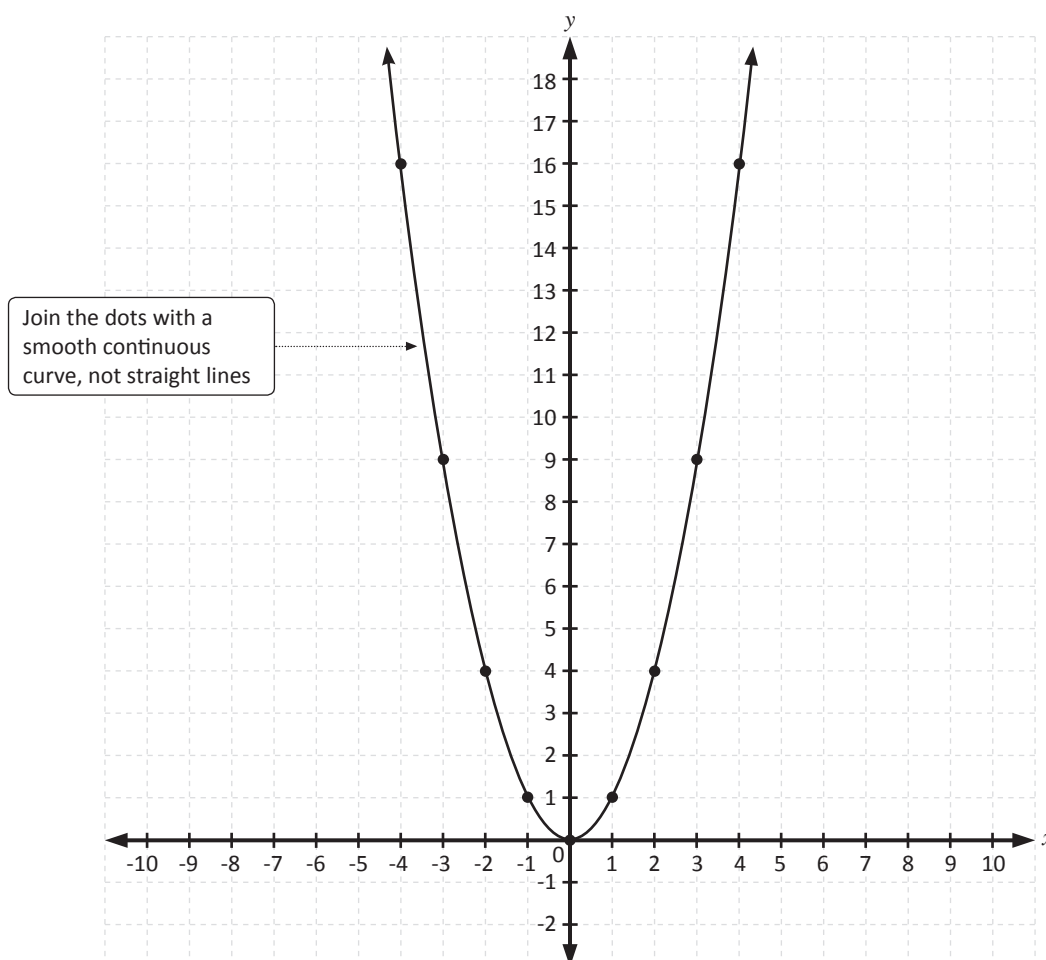
Using a Table to Plot a Parabola

Remember, the table method can be used for ANY relationship between y and x . Here's an example with $a = 1$, so $y = x^2$.

Complete this table of values for $y = x^2$ and use it to plot the graph on the grid below

x	-4	-3	-2	-1	0	1	2	3	4
y	16	9	4	1	0	1	4	9	16

Now plot the points (x,y) : $(-4,16), (-3,9), (-2,4), (-1,1), (0,0), (1,1), (2,4), (3,9), (4,16)$



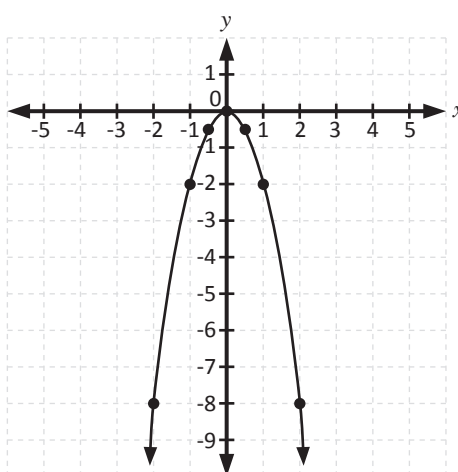
This shape  is called a 'Parabola'.

The Parabola Faces Downwards if a is Negative

Let's say a is negative, then the parabola will be 'upside down'. Here's an example with $a = -2$.

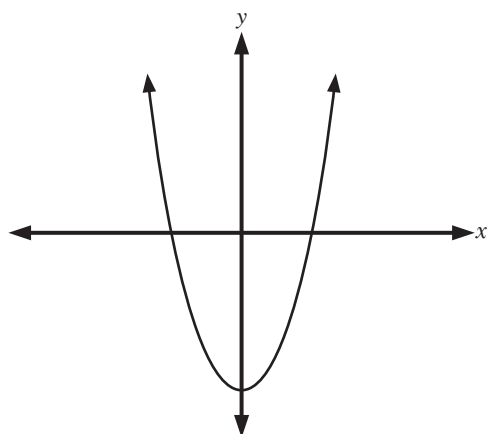
Complete this table of values for $y = -2x^2$ and use it to plot the graph on the grid below

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y	-8	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2	-8

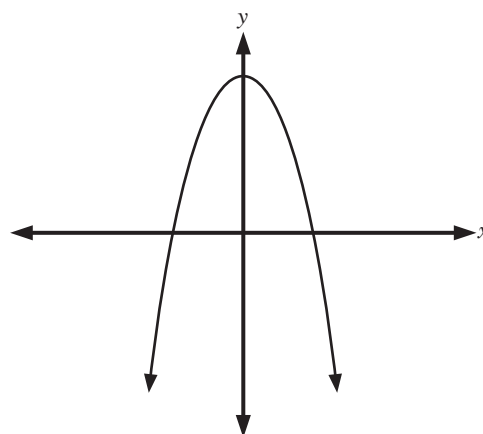


The **concavity** of a parabola is the direction it faces. It could be facing up or facing down.

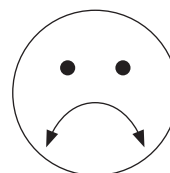
If a is positive ($a > 0$)
concave up



If a is negative ($a < 0$)
concave down



Sometimes parabolas which are concave upwards are called '**happy parabolas**' and parabolas which are concave downwards are called '**sad parabolas**'.



Parabolas of the Form $y = ax^2 + c$

What about a parabola for the functions like $y = 2x^2 + 2$ or $y = 2x^2 - 2$?

Complete the tables for these functions

$$y = 2x^2 - 2$$

x	-2	-1	0	1	2
y	6	0	-2	0	6

$$y = 2x^2$$

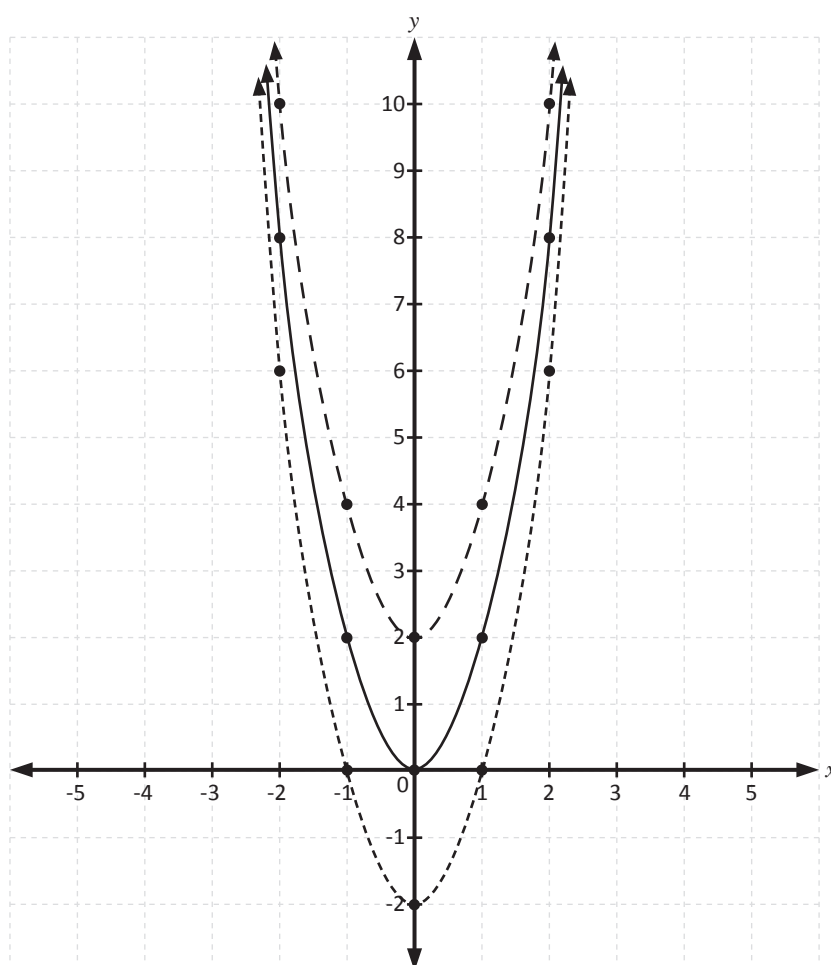
x	-2	-1	0	1	2
y	8	2	0	2	8

$$y = 2x^2 + 2$$

x	-2	-1	0	1	2
y	10	4	2	4	10

Use the tables to graph each function on the same set of axes

.....	$y = 2x^2 - 2$
———	$y = 2x^2$
---	$y = 2x^2 + 2$



The parabola for $y = 2x^2 + 2$ is the same as $y = 2x^2$, just shifted 2 units upwards.

The parabola for $y = 2x^2 - 2$ is the same as $y = 2x^2$, just shifted 2 units downwards.

The rule for these parabolas is:

- $y = ax^2 + c$ is shifted c units upwards of $y = ax^2$.
- $y = ax^2 - c$ is shifted c units downwards of $y = ax^2$.

1. Identify whether these relationships are linear or quadratic.

a $y = -x + 2$

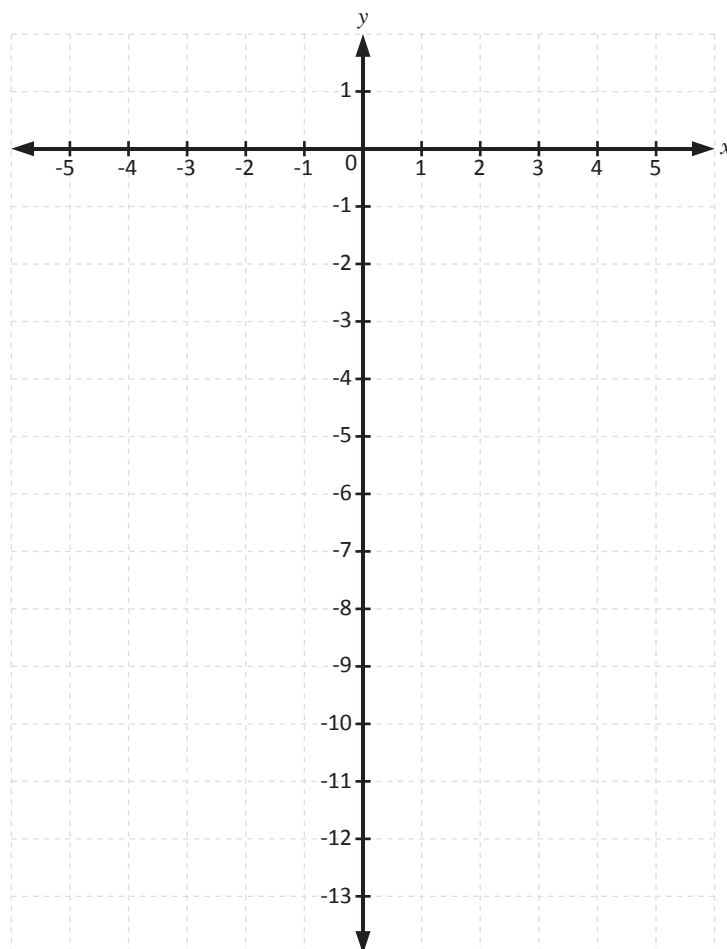
b $y = x^2$

c $y = \frac{-3}{2}x^2$

d $y = x + x^2$

2. Complete the table and draw the graph of $y = -3x^2$.

x	-2	-1	0	1	2
y					



Is this parabola concave upwards or concave downwards?

3. Answer these questions based on the function $y = \frac{1}{2}x^2$.

- a Complete this table based on the function.

x	-3	-2	-1	0	1	2	3
y							

- b Use the space below to draw your own set of axes and the graph of this function.

- c Is this parabola concave upwards or concave downwards?

4. Identify if the parabola for these functions is concave upwards or concave downwards (without drawing the graphs).

a $y = -7x^2$

b $y = \frac{x^2}{3}$

5. $y = \frac{1}{4}x^2$ is a quadratic function.

- a Complete this table for this function.

x	-4	-2	0	2	4
y					

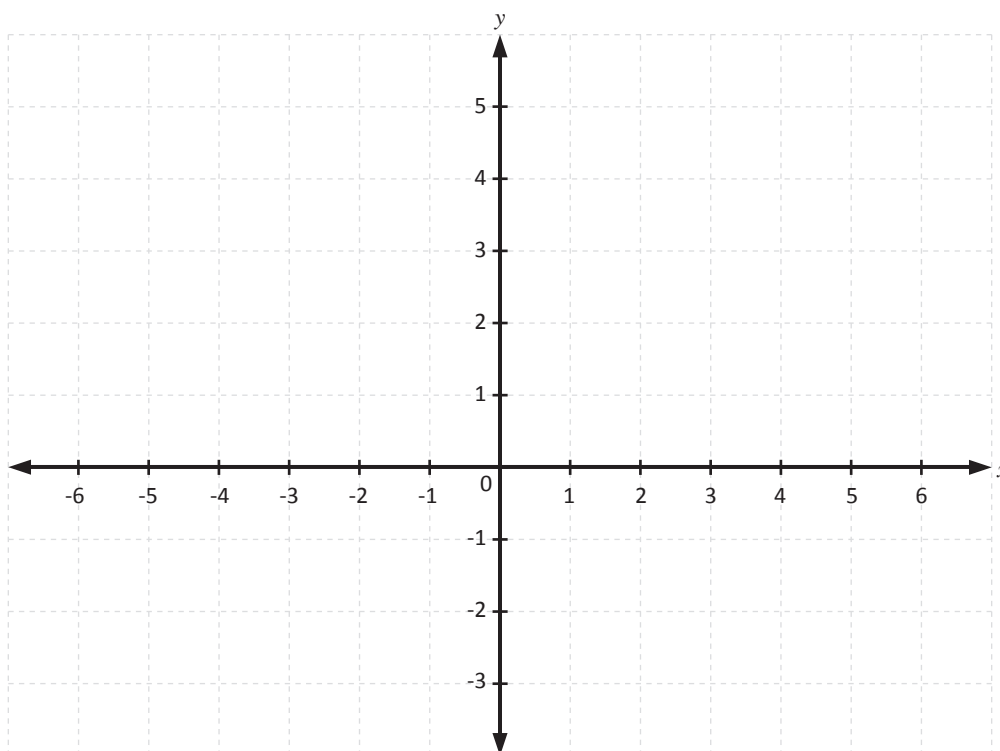
- b Complete the table for the function $y = \frac{1}{4}x^2 + 3$.

x	-4	-2	0	2	4
y					

- c Complete the table for the function $y = \frac{1}{4}x^2 - 3$.

x	-4	-2	0	2	4
y					

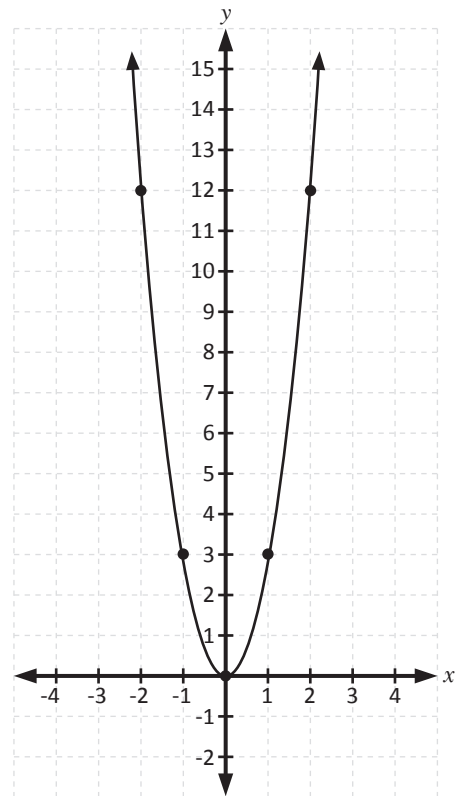
- d Use the tables plot all 3 graphs on this system of axes.



- e What do you notice about the graphs of $y = \frac{1}{4}x^2 + 1$ and $y = \frac{1}{4}x^2 - 1$ in function to $y = \frac{1}{4}x^2$?

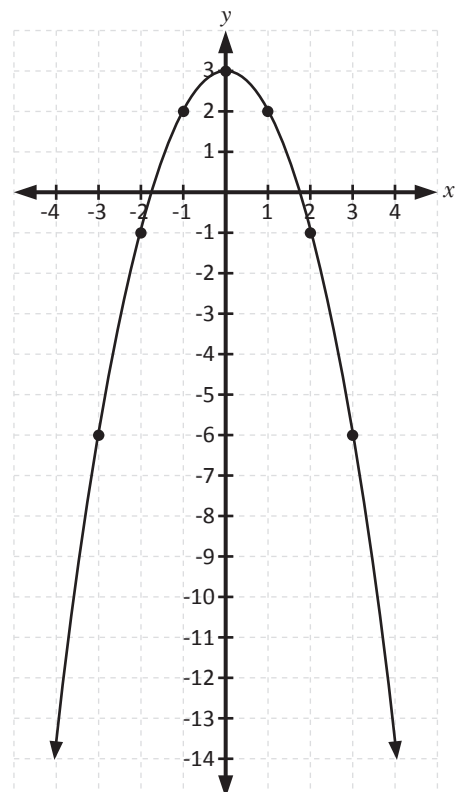
6. The graph on the right represents $y = 3x^2$.

- a On the same set of axes, draw the graph of $y = 3x^2 + 4$ without using a table.
- b On the same set of axes, draw the graph of $y = 3x^2 - 2$ without using a table.
- c Are these graphs concave upwards or concave downwards?
- d Does adding or subtracting a constant change the concavity?



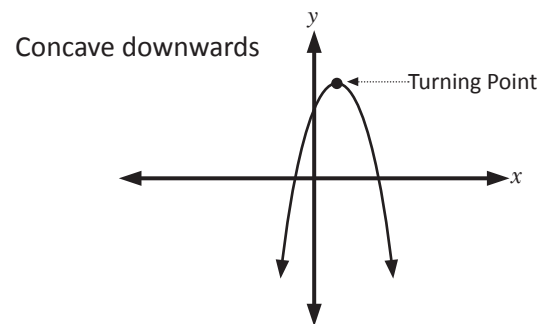
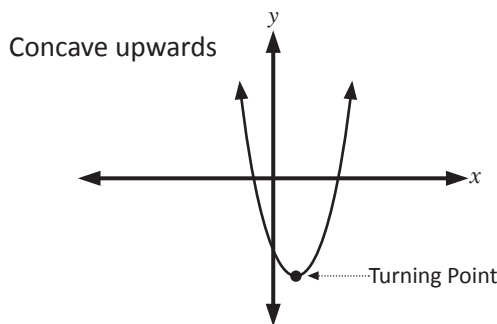
7. The graph on the right represents $y = -x^2 + 3$.

- a On the same set of axes, draw the graph of $y = -x^2$ without using a table.
- b On the same set of axes, draw the graph of $y = -x^2 - 6$ without using a table.
- c Are these graphs concave upwards or concave downwards?
- d Does adding or subtracting a constant change the concavity?



The Turning Point

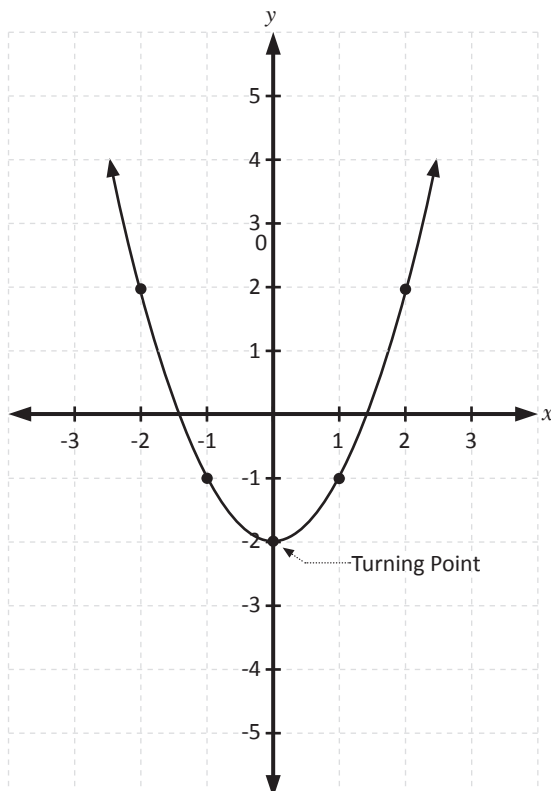
The turning point of the parabola is the coordinates of the point where the parabola 'turns around'.



Another name for the turning point is the 'vertex'. Here are some examples.

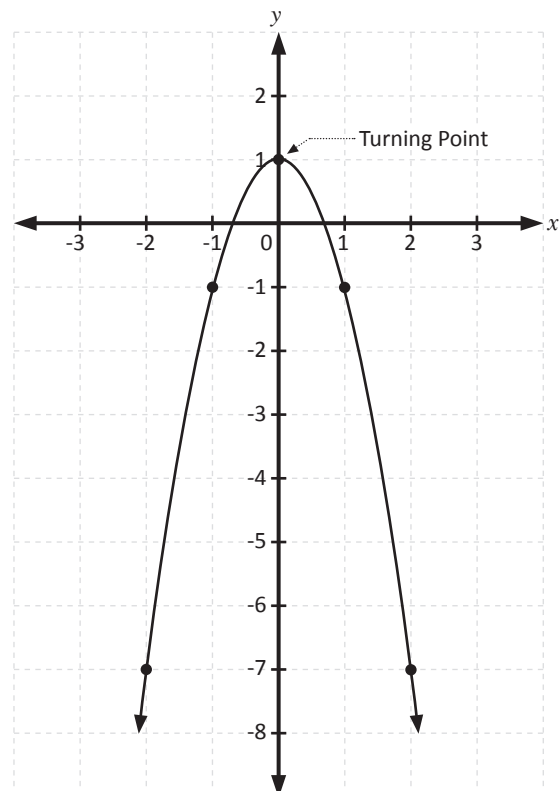
Find the turning points of the following

a $y = x^2 - 2$



The turning point or vertex of this parabola is $(0, -2)$

b $y = -2x^2 + 1$



The turning point or vertex of this parabola is $(0, 1)$

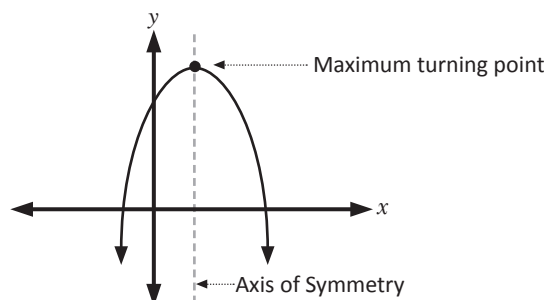
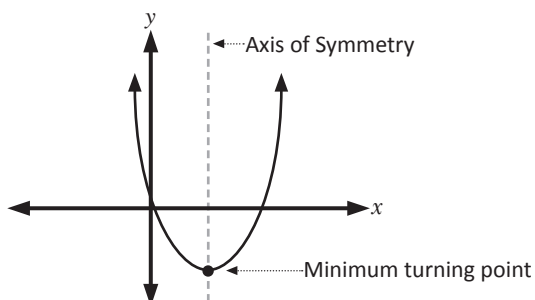
The vertex of a concave upwards parabola is at the lowest point and so it is called a **minimum turning point**.
The vertex of a concave downwards parabola is at the highest point and so it is called a **maximum turning point**.

This means that:

- If $a > 0$ (positive), then $y = ax^2 + c$ will have a **minimum** turning point.
- If $a < 0$ (negative), then $y = ax^2 + c$ will have a **maximum** turning point.

The Axis of Symmetry

The 'axis of symmetry' of a parabola is the vertical line which passes through the turning point.



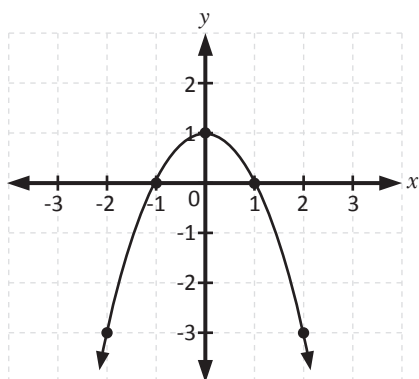
Since the axis of symmetry is a vertical line, it will have the equation

$$x = x\text{-value of turning point}$$

Find the axis of symmetry of each of these parabolas

a $y = -x^2 + 1$

x	-2	-1	0	1	2
y	-3	0	1	0	-3



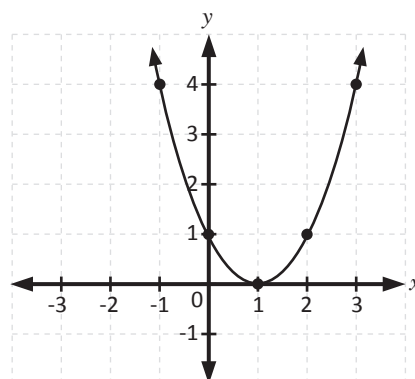
The turning point is $(0, 1)$

x -value

Axis of symmetry is $x = 0$

b $y = (x - 1)^2$

x	-1	0	1	2	3
y	4	1	0	1	4



The turning point is $(1, 0)$

x -value

Axis of symmetry is $x = 1$

Some important points about these examples

- In **a** the axis of symmetry is the y -axis or $x = 0$. In **b** the axis of symmetry is the vertical line $x = 1$.
- a** has a maximum turning point, and a maximum y -value of 1.
- b** has a minimum turning point, and a minimum y -value of 0.

1. Why do some parabolas have maximum turning points and others have minimum turning points?

2. Use a table to draw this parabola: $y = 4 - x^2$ on the axes provided. (Hint: $y = 4 - x^2$ is the same as $y = -x^2 + 4$)

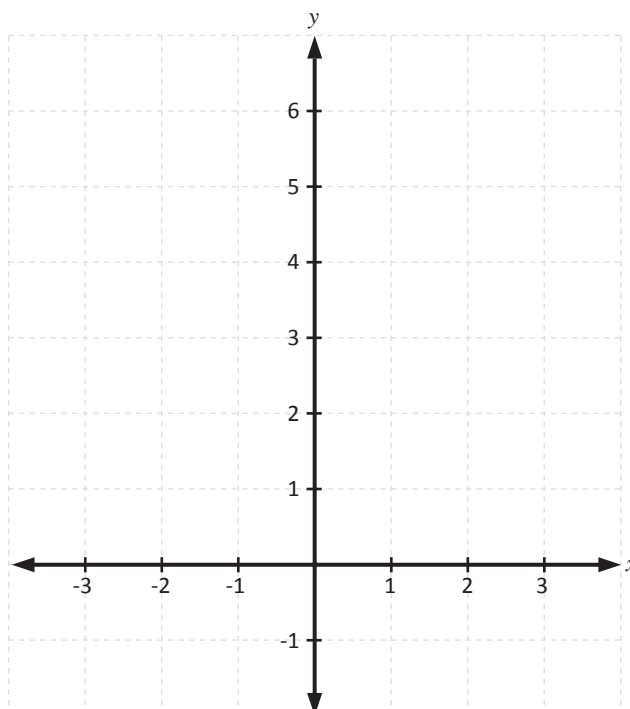
a

x	-2	-1	0	1	2
y					

b What is the vertex of this parabola?

c Is the vertex a maximum or a minimum?

d What is the axis of symmetry?



3. Without drawing anything, state whether each of the following will have a minimum or maximum turning point.

a $y = 2x^2 - 3$

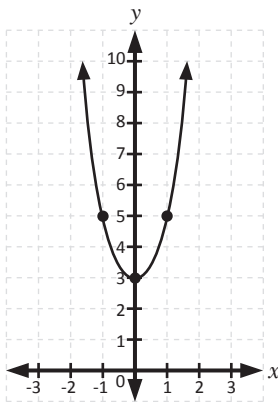
b $y = -3x^2 + 7$

c $y = 5 + 6x^2$

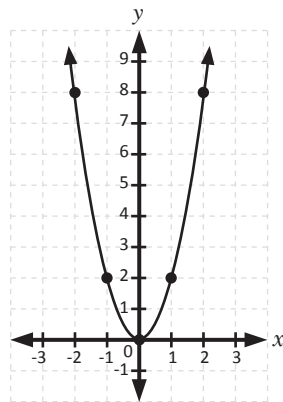
d $y = 5 - 5x^2$

4. Match these equations to the parabolas below.

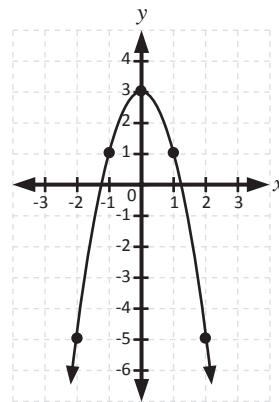
a $y = 2x^2$



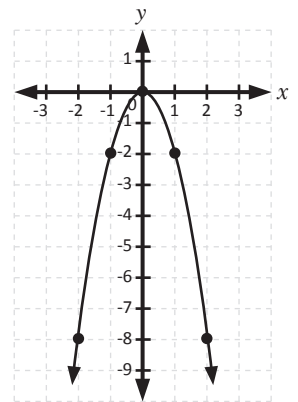
$y = -2x^2$



$y = 2x^2 + 3$



$y = 3 - 2x^2$



b Identify which equation has a maximum turning point at $(0, 0)$.

c Identify which equation has a minimum turning point at $(0, 0)$.

d What is the axis of symmetry of all these functions?

e Identify which equation has a maximum y -value of 3.

f Identify which equation has a minimum y -value of 3.

There are two more important nonlinear graphs called the 'hyperbola' and 'exponential'. The table method is used for both.

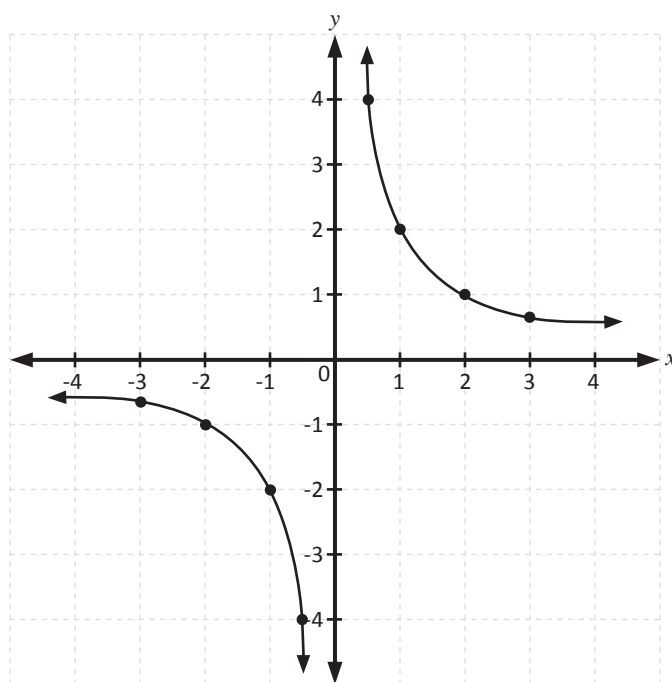
Using a Table to Plot the Hyperbola

A hyperbola is the graph of an equation $y = \frac{k}{x}$ where k can be any number like $y = \frac{2}{x}$ or $y = \frac{-3}{x}$.

Remember, the table method can be used for ANY function between y and x . Here's an example:

Complete this table of values for $y = \frac{2}{x}$ and use it to plot the graph on the grid below

x	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
y	$-\frac{2}{3}$	-1	-2	-4	UD	4	2	1	$\frac{2}{3}$



Four Important Things to Notice

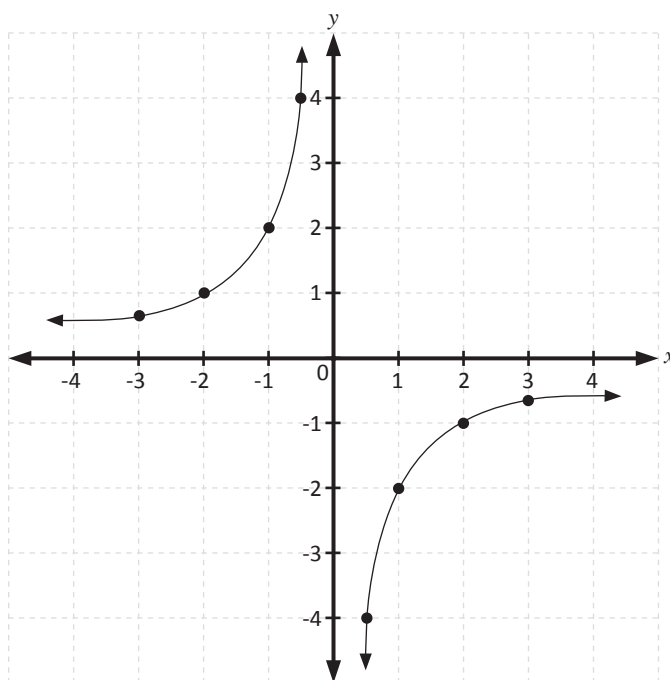
- Obviously x can never be 0 because anything divided by 0 is undefined. You can see that in the table.
- The x and y axis are **asymptotes**. This is a fancy way of saying that the graph gets closer and closer towards the x and y axis, but never touches them.
- The hyperbola is split into two separate **branches** in opposite quadrants (1st and 3rd). It is **not** continuous.
- The branches are the same size and symmetrical about the line $y = -x$.

What if k is Negative?

Let's say k is negative ($k < 0$). The Hyperbola will be in the other quadrants. Here's an example with $k = -2$.

Complete this table of values for $y = \frac{-2}{x}$ and use it to plot the graph on the grid below

x	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3
y	$\frac{2}{3}$	1	2	4	UD	-4	-2	-1	$-\frac{2}{3}$



As you can see, when $k < 0$ the branches of the hyperbola are in the 2nd and 4th quadrants

When k is negative the same properties hold as when k is positive. The only difference is the quadrants the branches are in.

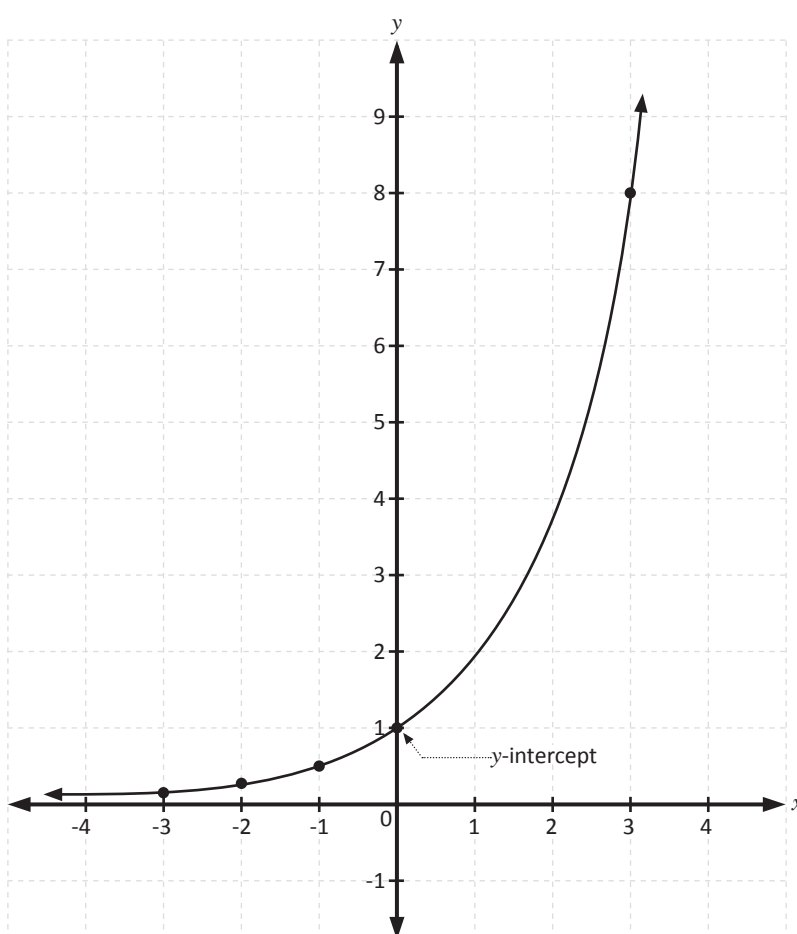
- Obviously x STILL can never be 0 because anything divided by 0 is undefined.
- The x and y axis are still **asymptotes**. The hyperbola never touches them.
- The hyperbola is split into two separate **branches** in opposite quadrants (2nd and 4th). It is **not** continuous.
- The branches are the same size and symmetrical about the line $y = x$.

Using a Table to Plot an Exponential

In an exponential function the x is in the exponent, $y = a^x$. a is any positive number. Some examples are $y = 3^x$ or $y = 2^{-2x}$. As usual, here is an example which uses the table method to plot an exponential graph.

Complete this table of values for $y = 2^x$ and use it to plot the graph on the grid below

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8



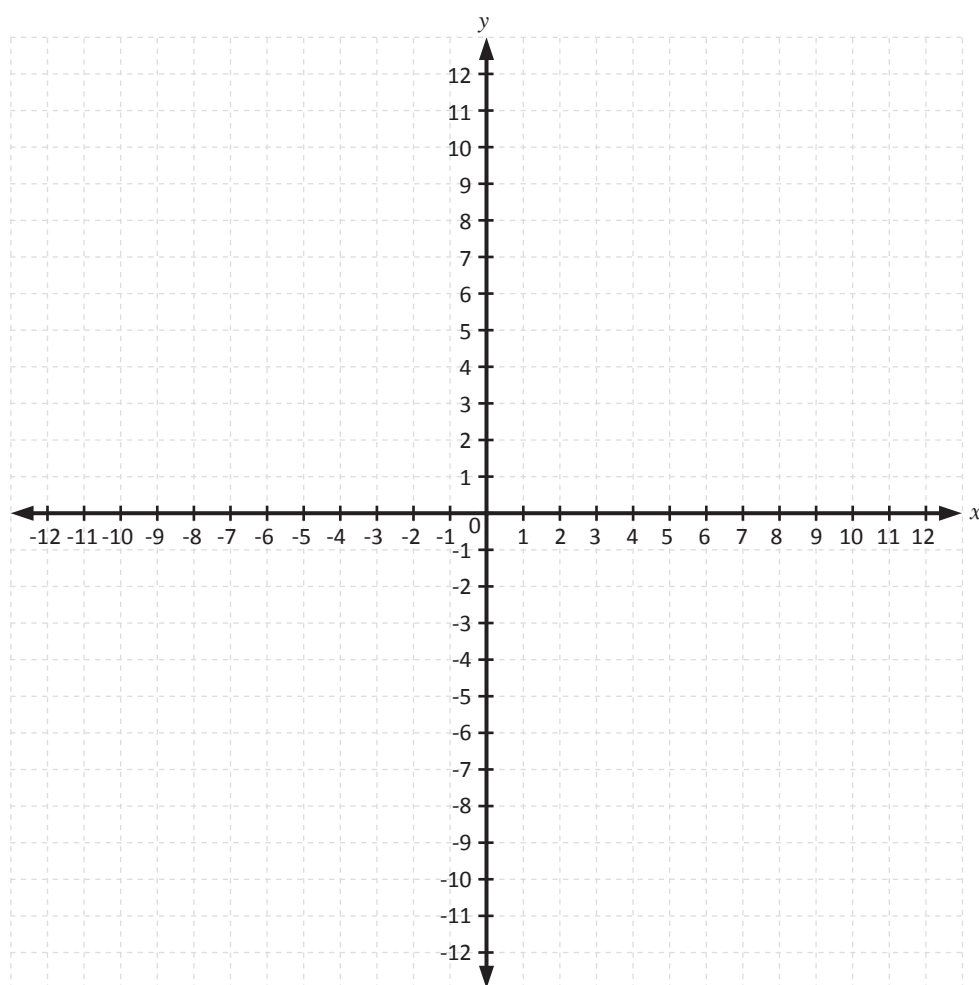
Important things to notice

- For any value of a in $y = a^x$, the y -intercept is **ALWAYS** at $(0, 1)$.
- The x -axis is an **asymptote**. The graph gets closer but does not touch it. The y -axis is not an asymptote, because the graph passes through it.
- The y -value increases as the x -value increases.
- The y -value decreases as the x -value decreases.

1. $y = \frac{12}{x}$ can be written as $xy = 12$.

a Use the table and axes below to draw the graph of $xy = 12$.

x	-6	-4	-3	-2	-1	1	2	3	4	6
y										

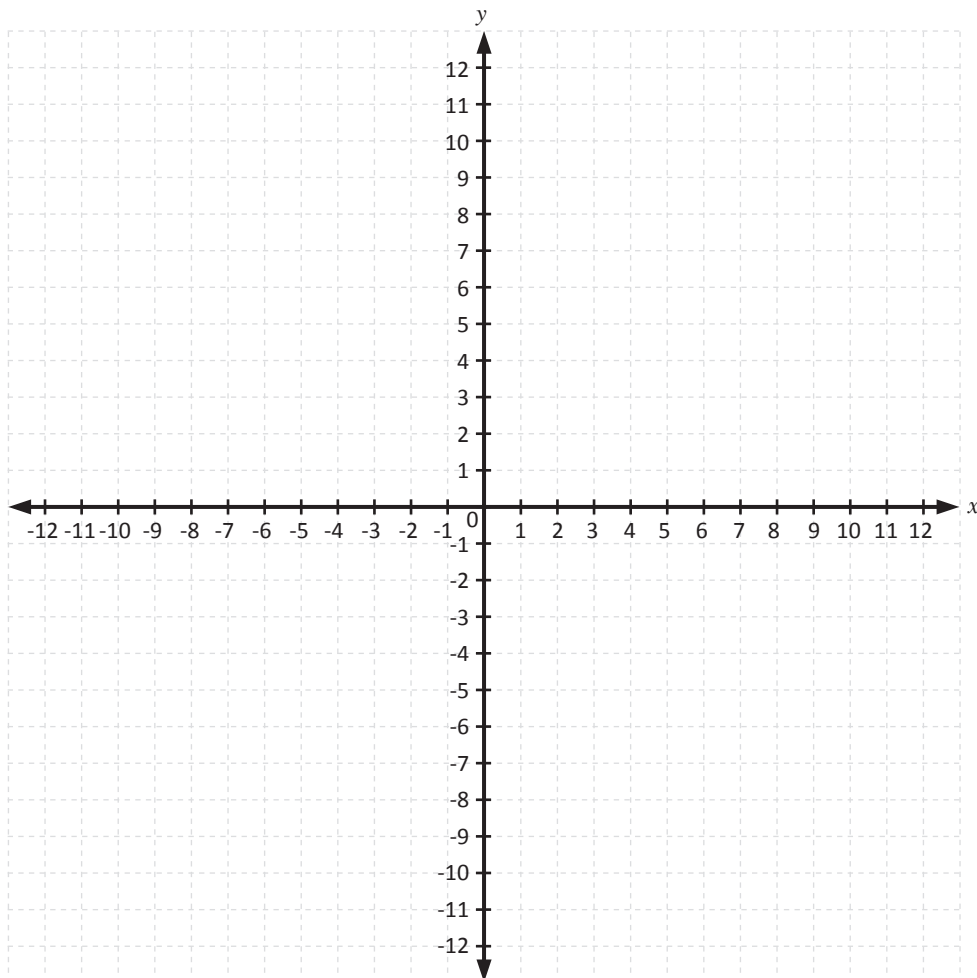


b Will either of the branches ever touch the x or y axis?

c Which quadrants are the branches of the hyperbola in?

- d Use the table and axes below to draw the graph of $xy = -12$.

x	-6	-4	-3	-2	-1	1	2	3	4	6
y										

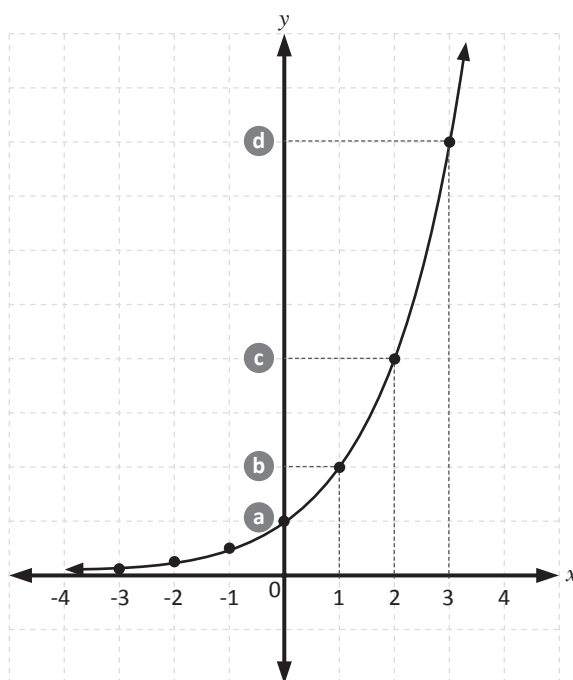


- e What is different in the above hyperbola from the hyperbola on the previous page?
- f Highlight the asymptotes on each of the above graphs.

2. The exponential equation is an equation of the form $y = a^x$ where a is any positive number.

- a What is the value of 5^0 ?
- b What is the value of a^0 where a is any positive number?
- c What is the y-intercept of any exponential equation $y = a^x$? Why?

3. Find the missing values on the y-axis, if the graph below is of the function $y = 4^x$.



a

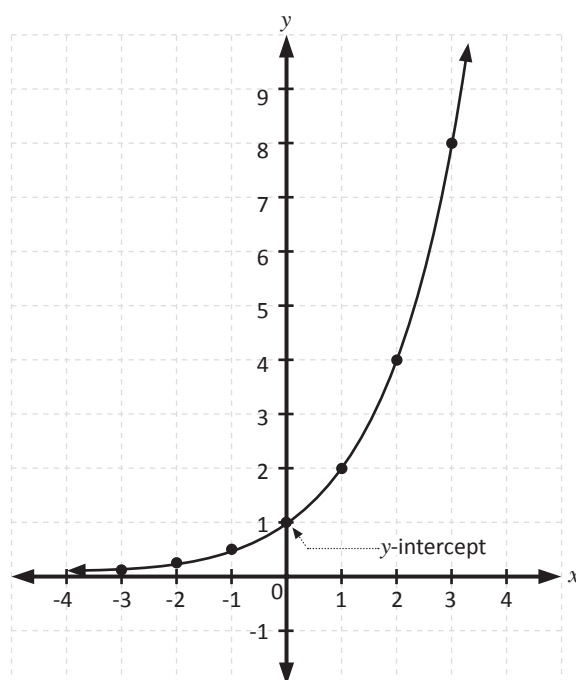
b

c

d

4. The table below has been completed for $y = 2^x$ and the graph plotted underneath.

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8



a Complete the table below for $y = 2^{-x}$.

x	-3	-2	-1	0	1	2	3
y							

b Plot the graph of $y = 2^{-x}$ on the same set of axes above.

c In the graph of $y = 2^x$, y increases as x increases. Is this the same in $y = 2^{-x}$?

d Highlight the asymptotes for both graphs.

Basics:

- a Linear b Nonlinear

c Linear d Linear

e Linear f Nonlinear

- a $y = x + 3$

x	-2	-1	0	1	2
y	1	2	3	4	5

- $y = 2x$

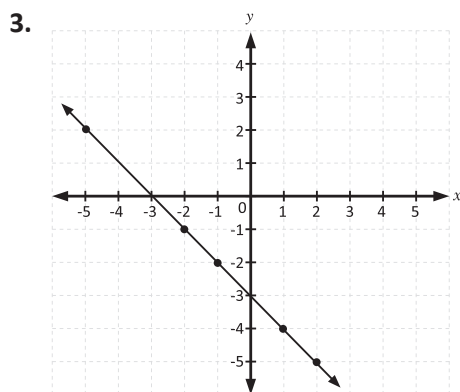
x	-2	-1	0	1	2
y	-4	-2	0	2	4

- $y = -x$

x	-2	-1	0	1	2
y	2	1	0	-1	-2

- $y = -x - 3$

x	-2	-1	0	1	2
y	-1	-2	-3	-4	-5

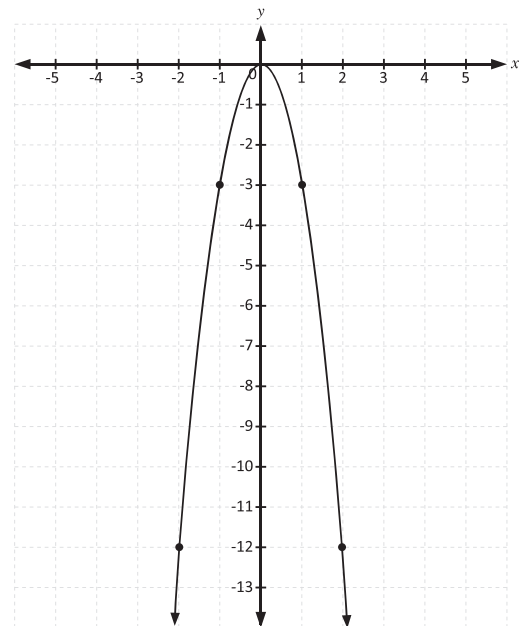


Knowing More:

- a Linear b Quadratic

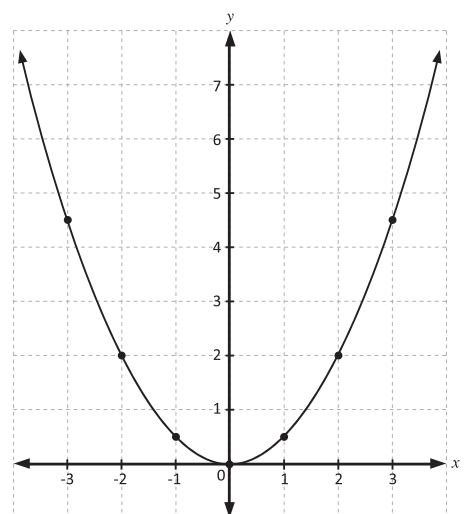
c Quadratic d Quadratic

- | x | -2 | -1 | 0 | 1 | 2 |
|---|-----|----|---|----|-----|
| y | -12 | -3 | 0 | -3 | -12 |



Parabola concave downward (sad)

- | x | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|---|-----|----|-----|---|-----|---|-----|
| y | 4.5 | 2 | 0.5 | 0 | 0.5 | 2 | 4.5 |



Parabola concave upwards (happy)

Knowing More:

4. **a** $y = -7x^2$ concave downward (sad)

b $y = \frac{x^2}{3}$ concave upwards (happy)

5. **a**

x	-4	-2	0	2	4
y	4	1	0	1	4

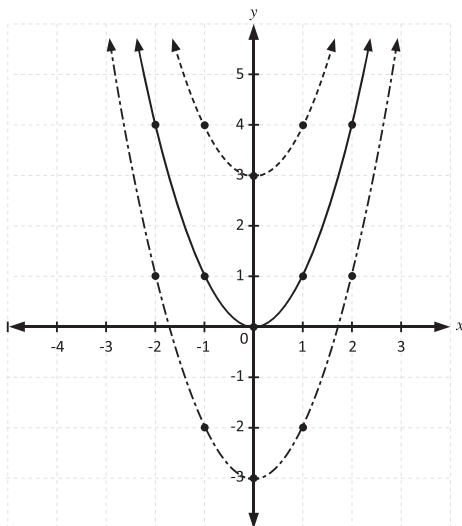
b

x	-4	-2	0	2	4
y	7	4	3	4	7

c

x	-4	-2	0	2	4
y	1	-2	-3	-2	1

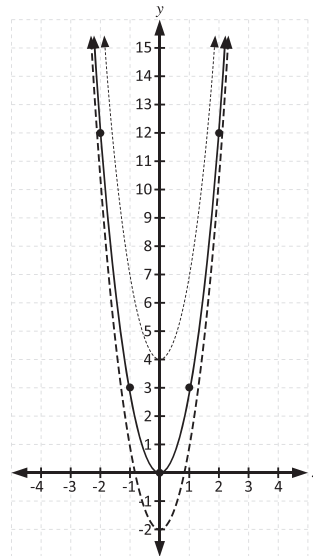
d



e When 3 is added the curve moves 3 units up the y-axis. Similarly, when 3 is subtracted the curve moves 3 units down the y-axis.

Knowing More:

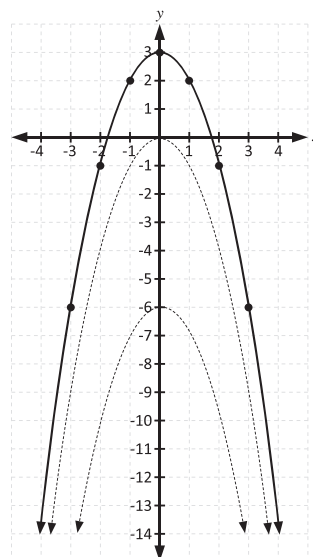
6. **a** and **b**



c All three graphs are concave upwards (happy)

d No. Adding or subtracting a constant does not change concavity

7. **a** and **b**



c All three graphs are concave downwards (sad)

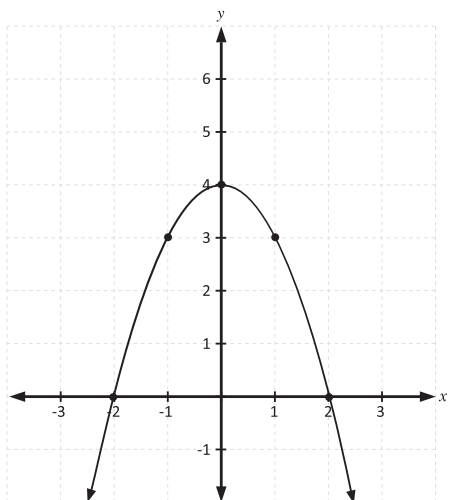
d No. Adding or subtracting a constant does not change concavity

Using Our Knowledge:

1. Concave upward parabolas (where $a > 0$) have a turning point that is a minimum. While concave downward parabolas (where $a < 0$) have a turning point that is a maximum.

2. a

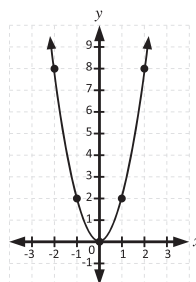
x	-2	-1	0	1	2
y	0	3	4	3	0



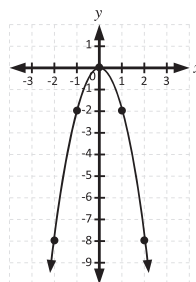
- b (0, 4)
- c Maximum. Concave downwards (sad) parabolas always have a vertex that is a maximum.
- d $x = 0$
3. a Minimum at (0, -3)
- b Maximum at (0, 7)
- c Minimum at (0, 5)
- d Maximum at (0, -5)

Using Our Knowledge:

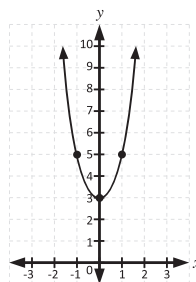
4. a $y = 2x^2$



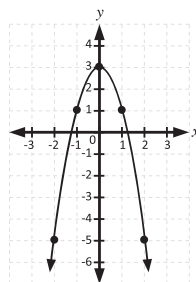
$y = -2x^2$



$y = 2x^2 + 3$



$y = 3 - 2x^2$

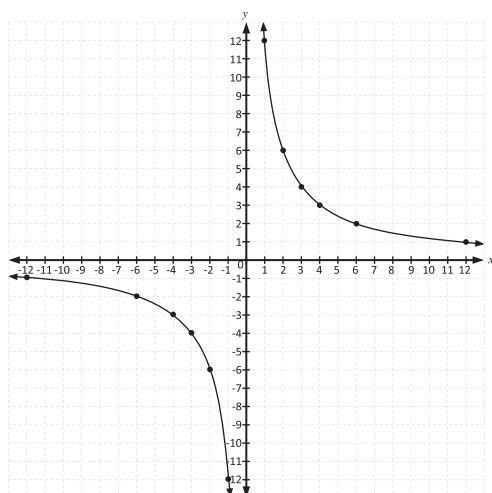


- b $y = -2x^2$
- c $y = 2x^2$
- d $x = 0$
- e $y = 3 - 2x^2$
- f $y = 2x^2 + 3$

Thinking More:

1. a

	6	2
	4	3
	3	4
	2	6
	1	12
	-1	-12
	-2	-6
	-3	-4
	-4	-3
	-6	-2
x	y	

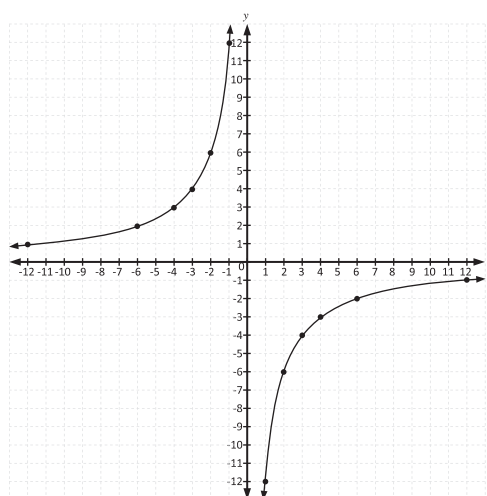


- b No. Both the x and y axes are asymptotes.
- c The hyperbola has a branch in each of the 1st and 3rd quadrants.

Thinking More:

1. d

	6	-2
	4	-3
	3	-4
	2	-6
	1	-12
	-1	12
	-2	6
	-3	4
	-4	3
	-6	2
x	y	



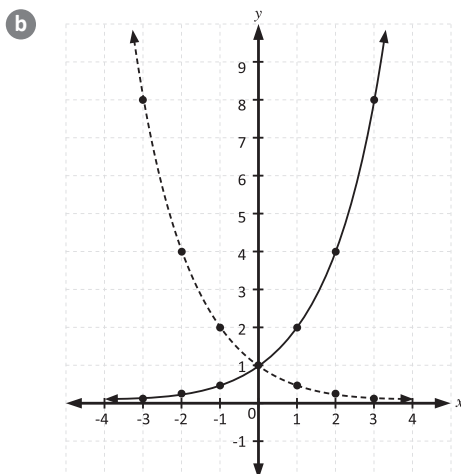
- e This hyperbola has a branch in each of the 2nd and 4th quadrants (rather than 1st and 3rd)
- f The asymptotes for both of these hyperbolas is the x and y axes. This means that the graph will get very close to these asymptotes but will never touch these.

Thinking More:

2. **a** The value of $5^0 = 1$
- b** The value of $a^0 = 1$
- c** The y-intercept is always 1 for any exponential equation of the form $y = a^x$. This is because any number raised to the power of zero will be one.
-
3. **a** $(0, 1) \therefore y = 1$ **b** $(1, 4) \therefore y = 4$
- c** $(2, 16) \therefore y = 16$ **d** $(3, 64) \therefore y = 64$
-

4. **a**

x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$



- c** No. As x increases, y decreases. The equation $y = 2^{-x}$ is shown in the above graph using the dashed line.
- d** The asymptote for both graphs is the same. It is the x -axis or the line $y = 0$. In other words, while both the curves $y = 2^x$ and $y = 2^{-x}$ get very close to the x -axis, they never touch it.



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